

Battery System

ESS-LB16-W02



Overview

The low-voltage battery system includes a battery unit ESS-LB16-W02. It can be used in conjunction with energy storage inverters or photovoltaic energy storage integrated machines for storing and releasing electrical energy.

Features

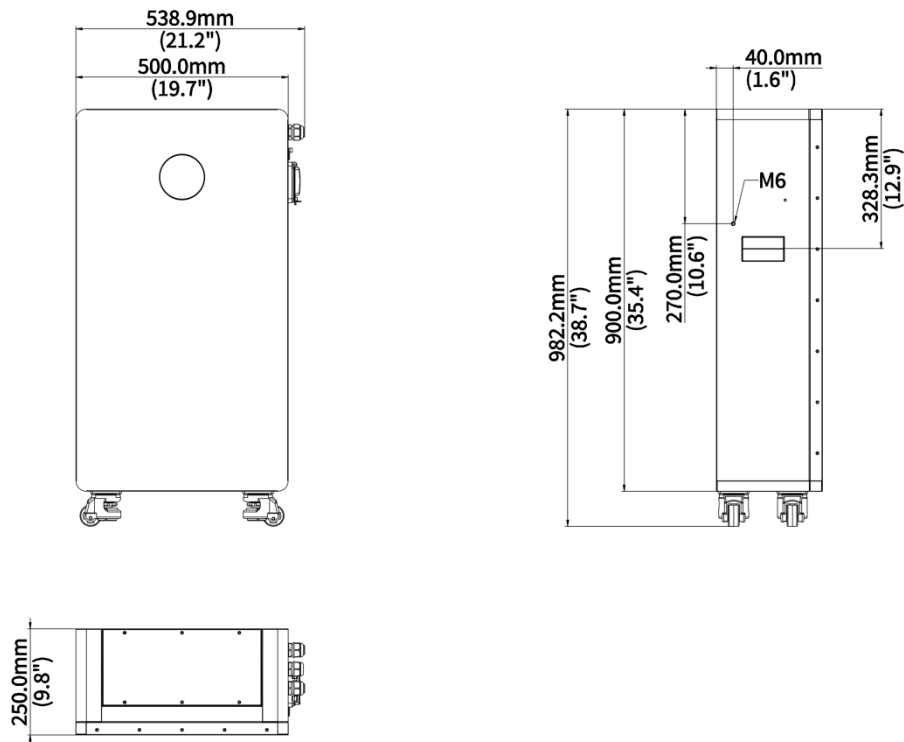
- Utilizes lithium iron phosphate batteries, providing high safety and long lifespan.
- Equipped with a self-developed battery management system with protective functions like over-discharge, overcharge, overcurrent, and temperature protection.
- Employs high-precision algorithms for SOC (State of Charge) and SOH (State of Health) estimation, ensuring more accurate calculations.
- Modular design, supports 1 to 16 expansions, adaptive control strategies. Suited to residential and commercial applications for increasing the self-consumption ratio.
- Industrial-grade wide temperature range to withstand high and low temperature environments.
- Automotive-grade anti-interference design to withstand strong and complex electromagnetic environments.
- Equipped with fuses for electrical safety protection.
- IP65 protection rating makes it ready for both indoor and outdoor use.
- Standard configuration with caster design, which can be moved conveniently.
- Multiple safety certifications: UN38.3, MSDS, IEC62619, EN IEC61000, IEC62040 etc.
- Abundant and comprehensive fault diagnosis and protection functions provide timely protection and accurate troubleshooting.

Specifications

Model	ESS-LB16-W02
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BOM	
BOM	0235CBHY
Electrical	
Cell Type	LFP (LiFePO4)
Rated Energy	16.08 kWh
Rated Capacity	314 Ah
Rated Voltage	51.2 V
Voltage Range	43.2 V to 58.4 V
Rated Charge Current	160 A
Max Charge Current	200 A
Rated Discharge Current	160 A
Max Discharge Current	200 A
Cycle Life	≥ 8000 (25±2 °C, 90%DOD, 0.5 C/0.5 C, 70%EOL)
Communication	CAN, RS485
Basic Parameters	
Battery Unit Dimensions	500 (W)*250 (D)*900 (H) mm (excluding casters)
Battery Unit Weight	120 kg
Cooling	Passive cooling
Certificates	UN38.3 / MSDS / IEC62619 / EN IEC 61000-6-1 / EN IEC 61000-6-3 / IEC62040 etc.
Ingress Protection	IP65
Installation Style	Ground standing mount
Operating Environment	
Operating Temperature	Charging: 0 °C to 55 °C Discharging: -30 °C to 55 °C
Storage Temperature	-20 °C to 60 °C
Operating Humidity	20% to 90%, RH (non-condensing)
Altitude	≤ 4000 m
Accessories	
Inverter Cable Package	PCS communication cable×1

Dimensions



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Product specifications and availability are subject to change without notice.